Keshav Bagri

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Education

The Ohio State University — **Columbus, OH**

*Master of Science in Mechanical Engineering [GPA: **4.0/4.0**] | Advisor: Prof. Giorgio Rizzoni* — *2022 – 2024*

Graduate Specialization in Automotive Systems and Mobility

***Thesis**: Quantitative risk assessment and mitigation through fault diagnostics for automated vehicles [Link]*

Indian Institute of Technology (IIT) Kharagpur — **Kharagpur, India**

*Bachelor of Technology (Hons.) in Mechanical Engineering [CGPA: **9.26/10**]* — *2018 – 2022*

*Micro Specialization in Embedded Control, Software, Modelling and Design [CGPA: **9.5/10**]*

***Thesis**: Fault tolerant control system for Electric Vehicles*

Publications

- [Under review] **Bagri, K.**, Rizzoni, G., "Risk quantification and mitigation through fault diagnostics for highly automated vehicles" in IEEE Transactions on Intelligent Vehicles.
- **Bagri, Keshav.** "Quantitative risk assessment and mitigation through fault diagnostics for automated vehicles." Master's thesis, Ohio State University, 2024. [Link]
- Singh, Y., **Bagri, K.**, Jayakumar, A., Rizzoni, G. Fault Diagnostics for Oscillatory Failure Case in Aircraft Elevator Servos. IFAC World Congress in Yokohama, Japan, July 2023. [Link] [Media]

Professional affiliations

- IEEE Transportation & Electrification Conference (ITEC)
  - Short Courses/Tutorials and Track Chair [*2026*]
  - Session chair for Fast and Onboard Charging [*2025*]
- As a Member
  - ASME Technical Committee on Automotive and Transportation Systems [*2025*]
  - IEEE Control Systems Society TCs on System Identification & Adaptive Control and Automotive Controls [*2025*]
  - IFAC TCs on Automotive Control (TC 7.1) and Transportation Systems (TC 7.4) [*2025*]
  - SAE Dynamical Modeling & Simulation and Hybrid EV Committees [*2025*]
  - SAE J3271 Megawatt Charging System for Electric Vehicles Task Force [*2025*]
  - SAE J3341 Carbon Footprint Methodologies Passenger Vehicles Task Force [*2025*]
- As a Reviewer
  - ASME Journal of Dynamic Systems, Measurement and Control [*2025*]
  - IEEE Open Journal of Control Systems [*2025*]
  - IEEE Transactions on Transportation Electrification [*2025*]
  - SAE International Journal of Electrified Vehicles [*2025*]
  - SAE International Journal of Sustainable Transportation, Energy, Environment, & Policy [*2025*]
  - SAE International Journal of Passenger Vehicle Systems [*2025*]
  - American Control Conference (ACC), Modeling Estimation and Control Conference (MECC) [*2024, 2025*]
  - IEEE Transportation & Electrification Conference (ITEC) [*2026*]

Awards and Achievements

- 2nd **place** in "Aerospace Competition on Fault Detection & Fault Tolerance" at **IFAC World Congress 2023** by **Airbus**
- **Gold medal** in Inter IIT Tech Meet 10.0 (2022) in the event "Powered Bonnet for EVs" by **Jaguar Land Rover India**
- **1st place** in Combustion category & Engineering Design among **31 teams** globally at Formula Bharat Virtuals 2021-22
- **Gold medal** in Inter IIT Tech Meet 9.0 (2021) in the event "Bosch's Electric Vehicle Simulation" by **Robert Bosch India**

Professional Experience

Lucid USA, Inc. — **Newark, CA**

Engineer, Battery Algorithms — *August 2024 – Present*

- Developing and refining algorithms for controlling the battery's performance with a focus on optimizing powertrain demands, enhancing user experience, and maximizing battery longevity
- Planning and executing comprehensive testing to verify software performance under extreme conditions, including temperature variations and unusual driving scenarios
- Collaborating with data analysts to analyze fleet and customer data, translating insights to improve estimation and control algorithms for charging

The MathWorks Inc. **Natick, MA**
Software Developer, Engineering Development Group | [Simulink Fault Analyzer] *May 2023 – August 2023*

- Engineered the **full stack development** of a feature to analyze fault trees, reducing safety engineers' workload by **50%**
- Incorporated functionalities to enable the precise determination of **failure probability** & identification of **minimal cut sets**
- Recognized limitations in the existing implementation, proposing actionable solutions to improve the efficiency by **30%**

Center for Automotive Research, The Ohio State University **Columbus, OH**
General Motors - SAE AutoDrive Challenge II | [Center for Automotive Research]

Graduate Research Associate *August 2023 – August 2024*

- Developed an optimal graph-based dynamic rerouting algorithm, enabling real-time on-demand computation of a new trajectory for obstacle avoidance in an urban driving environment, ensuring efficient navigation towards the global goal
- Devised a Finite State Machine - based behavior planner for navigation in an urban environment using object and traffic sign detections for decision-making and collision avoidance
- Coordinated with the systems safety, perception, & CAN teams to develop testing scenarios, considering key requirements

Co-Lead, Planning & Controls team *August 2022 – May 2023*

- Devised a polynomial-interpolation-based **trajectory generation** module to sample waypoints for lane change maneuver
- Formulated the logic for **lane changing maneuver** in a highway environment & performed SIL testing for edge cases
- Assisted in formulating the **Functional Interface Analysis** & Requirements Traceability Matrix, ensuring safety compliance

WMG, University of Warwick **Coventry, UK**
Research Collaborator | [Intelligent Vehicles] *January 2022 – July 2022*

- **Project**: Robust translation between 2 levels of Scenario Description Language
- Developed the Java-based translation framework between levels of **Scenario Description Language** for ADS simulation
- Formulated the mapping between environmental features and language variables using the taxonomy defined in **PAS 1883**
- Defined the boundary conditions for the agents' maneuver to describe the **Synchronised Serial Maneuver Sequences**

Revolute Robotics **Tucson, AZ**
Controls Engineer | [Revolute Robotics] *March 2022 – July 2022*

- **Project**: Develop optimal motion control algorithms for a Hybrid Mobility Robot
- Modified the **min. jerk trajectory** planner for smooth navigation using a dynamic window of **3 waypoints** per time step
- Formulated the cost-based **path planning & obstacle avoidance** approach using min. jerk planner & A* search algorithm
- Identified multiple methods to enable **data-logging** over cloud from the flight controller thus reducing manual intervention

Center for Automotive Research, The Ohio State University **Columbus, OH**
Research Intern | Advisor: Dr. Jeffrey P. Chrstos *February 2021 – March 2022*

- **Project**: Performance assessment of Driver-in-Loop simulators
- Reviewed literature for absolute and relative validity of medium fidelity D-i-L simulators for **physical validation**
- Proposed the **testing methodology** for the simulator's platform's motion and analyzed the results in the essence of OMCT
- Utilized the output of **LVDTs** for multiple displacement inputs to compute the platform's roll & pitch angles using MATLAB

Ati Motors **Bengaluru, India**
Winter Intern | Autonomy team *November 2020 – January 2021*

- **Project**: SLAM for autonomous cargo delivery vehicles
- Explored multiple **SLAM algorithms** like IMLS, EKF, Gmapping, etc. to select an efficient method for implementation
- Implemented the **Particle Filter** SLAM algorithm to build maps on a 2D grid using lidar and wheel odometry in Python
- Analysed the effects of multiple **resampling algorithms** and variation in **hyperparameters** on the map's quality

Projects

Formula SAE powertrain development | TeamKART | [Website] **IIT Kharagpur**
Supervisor: Prof. Dhananjay Kumar Srivastava *June 2019 – August 2021*

- Designed the **fuel tank** and the mounting arrangement for the vehicle considering the optimal capacity requirement
- Designed and analyzed the components of the **transmission** and **cooling system** for a standard FSAE electric vehicle
- Engineered the **intake manifold** & **crossflow radiator** for a single cylinder engine, producing a power output of **35 HP**
- Performed **engine simulations** in Ricardo WAVE and **CFD analysis** using Ansys Fluent for designing the intake manifold

Fault tolerant control system for electric vehicles **IIT Kharagpur**
Supervisor: Prof. Somnath Sengupta | Advanced Technology Development Center *February 2021 – April 2022*

- Worked on the modeling of integrated **ABS & regenerative braking** for efficient braking & maximum energy recuperation
- Engineered a **non-linear state estimator** using vehicle dynamics' equations to generate estimates of the vehicle's conditions
- Designed a novel **constraint-aware PI - sliding mode controller** for regulating the stability under all driving conditions
- Explored the possibility of integrating the novel controller with **reconfigurable control allocator** for fault-tolerant control

Mathematical modelling of Li-ion batteries focusing on Si anode particles **IIT Kharagpur**
Supervisor: Prof. Jeevanjyoti Chakraborty | Mechanical Engineering Department *February 2020 – January 2021*

- Studied the mechanics behind **crack development** and formation of amorphous lithiated Si with time around the Si anode
- Solved Ordinary and Partial differential equations, using **Finite Difference & Liebmann's methods**, in Python
- Established a **two-way coupling** relation between diffusion & stress, to understand the effect of one parameter on the other

Deep reinforcement learning for autonomous vehicles **IIT Kharagpur**
[Description] | Github *August 2020 – September 2020*

- Deployed a **Dueling DNN** to predict discrete action values & mapped them to continuous signals for vehicle control
- Used **84x84 RGB** images for the environment perception as the state information collected from the camera sensor
- Accommodated the wheel odometry, collision & lane invasion sensor data to compute **real-time reward** for the agent

Coursework / Technical Skills

Programming: C/C++, Git, Python, MATLAB, LaTex, Arduino, ROS, Atmel Studio
Softwares: MATLAB/Simulink, CARLA Simulator, Gazebo, COMSOL
Coursework: Powertrain control, Autonomy in Vehicles, Fault diagnosis, Vehicle dynamics & control, Linear Systems Theory

Teaching & Volunteering Experience

College of Engineering **The Ohio State University**
GUIDE Peer Mentor *August 2023 – April 2024*

- Responsible for assisting fresh graduate students in navigating the university by guiding them about university resources

Department of Mechanical & Aerospace Engineering **The Ohio State University**
Graduate Teaching Associate *January 2023 – May 2023*

- Courses: ME 3751 (Kinematics & Mechanism Design) and ME 3670 (Design & Analysis of Machine Elements)
- Responsible for conducting weekly recitations, office hours and doubt clarification sessions for a batch of 60+ UG students

Leadership Experience

Captain **The Ohio State University**
General Motors – SAE AutoDrive Challenge II *May 2023 – August 2024*

- Led a team of 20+ students from Ohio State to prototype the hardware & software stack for SAE Level 4 autonomy

Deputy Team Leader **IIT Kharagpur**
TeamKART, Formula SAE *July 2020 – August 2021*

- Spearheaded a dedicated team of **47** students towards the research & development of Formula Student prototype vehicle

Professional references

- Prof. Giorgio Rizzoni (Email: rizzoni.1@osu.edu)
  - Professor, Departments of MAE and ECE at The Ohio State University
  - Director at the Center for Automotive Research (OSU)
  - Ford Motor Company Chair in Electromechanical Systems
- Prof. Qadeer Ahmed (Email: ahmed.358@osu.edu)
  - Assistant Professor, Department of MAE at the Ohio State University
  - Core affiliate faculty of the Center for Automotive Research (OSU)
- Prof. Lisa Fiorentini (Email: fiorentini.2@osu.edu)
  - Clinical Professor, Department of ECE at the Ohio State University
  - Associate Fellow at the Center for Automotive Research (OSU)
- Prof. Andrea Serrani (Email: serrani.1@osu.edu)
  - Professor, Department of ECE at the Ohio State University
- Prof. Somnath Sengupta (Email: sengupta.s@atdc.iitkgp.ac.in)
  - Assistant Professor, Advanced Technology Development Center (ATDC) at IIT Kharagpur
- Ying Shi (LinkedIn)
  - Sr. Manager, Battery Algorithms at Lucid USA, Inc.
- Mahesh Nanjundappa (LinkedIn)
  - Software Engineering Manager at The MathWorks Inc.